HOW THE BUSINESS IS BUILT

Inside the micro-SaaS: the blocks, the flows, the money

Everything you need to understand to build, explain and sell a freight calculator and predictor. No code required.

1 Two machines
2 The 8 blocks
3 How it runs
4 The money machine
5 Build in 3 stages
6 Who does what

1 A micro-SaaS is two machines

A micro-SaaS is a small online software product, run by one person or a tiny team, that customers pay for every month. Yours has two parts that must both work.

MACHINE 1

The product machine

The technology. It takes a shipment in and gives answers out: total cost, when to book, whether the quote is fair, whether the deal is safe.

Question it answers: does it work?

MACHINE 2

The money machine

The business. It turns a stranger who reads your LinkedIn post into a user, and a user into someone who pays every month.

Question it answers: does it pay?

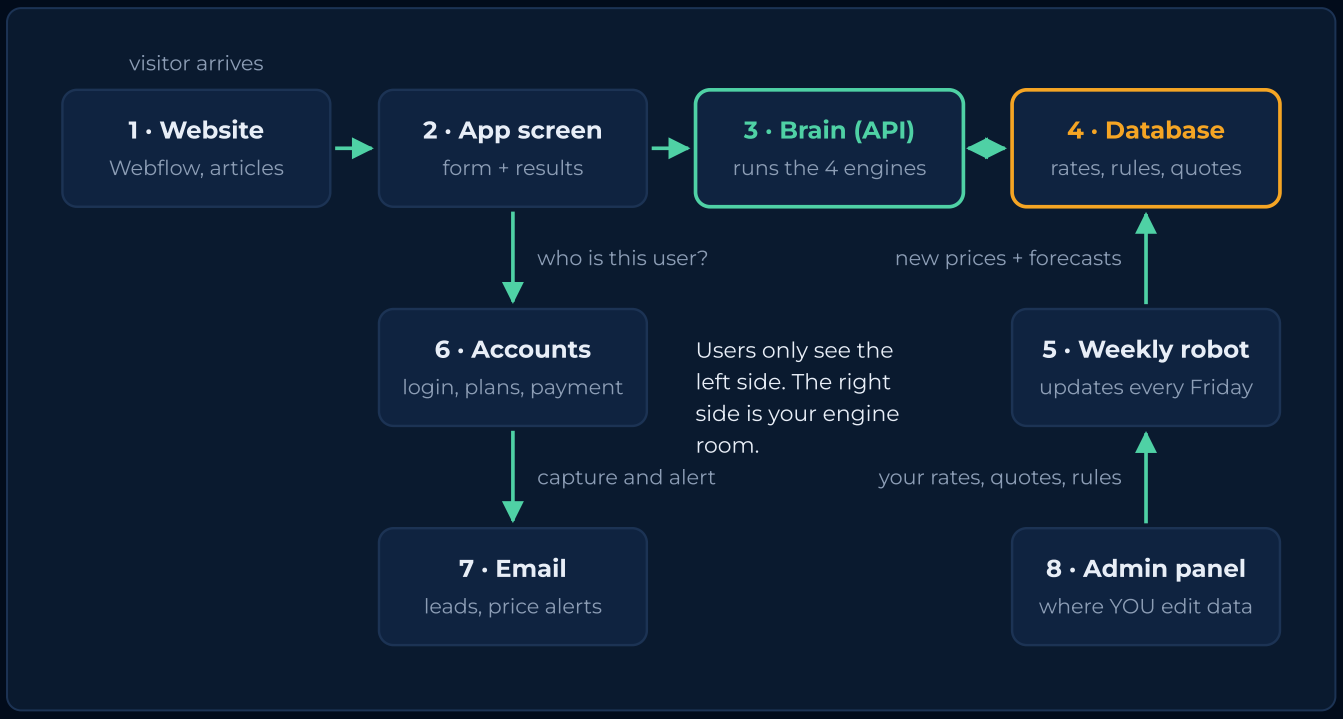
THE ONE IDEA TO REMEMBER

> **The code can be copied. Your data and your rules cannot. That is the business.**

Every design choice in this guide protects two things: the real Algeria freight data you collect, and the risk rules that come from your field experience.

2 The product machine: 8 blocks

Each block does one job. Arrows show what talks to what.



BLOCK	WHAT IT IS	THINK OF IT AS
1 · Website	Your Webflow site: landing page and articles	The shop window
2 · App screen	The form and the results the user sees	The order counter
3 · Brain (API)	A server that runs engines A, B, C, D	The kitchen
4 · Database	Freight rates, duty bands, risk rules, users, saved calculations, quotes	The fridge and the recipe book
5 · Weekly robot	A script that updates prices and forecasts every Friday	The weekly delivery
6 · Accounts	Login, free vs paid plans, the paywall, payment	The cash register
7 · Email	Collects emails, sends "your route moved" alerts	The loyalty card
8 · Admin panel	Private page to enter rates, quotes and new rules	The manager's office

WHY THE ENGINES RUN ON A SERVER, NOT ON THE PAGE

On the page (browser)

Anyone can open the page code and copy your formulas, rates and rules in two minutes. Fine for a free lead magnet. Not fine for a paid product.

On a server (API)

Users only see the answers. You update data once and everyone gets it. You can see what people calculate, which tells you what to build next.

3 How it runs: two flows

The product does two things on repeat: it answers users all day, and it updates itself once a week.

FLOW 1 · EVERY CALCULATION

1. **User fills the form**
   Route, container, value, Incoterm, payment, quote.
2. **Screen sends it to the brain**
   Only the inputs travel, never your formulas.
3. **Brain reads the database**
   Current rate, forecast, duty band, risk rules.
4. **Runs the 4 engines**
   Cost, forecast, quote check, risk.
5. **Sends answers back**
   The screen shows them in under a second.
6. **Saves the calculation**
   Anonymised. **Every use grows your data.**

FLOW 2 · EVERY FRIDAY

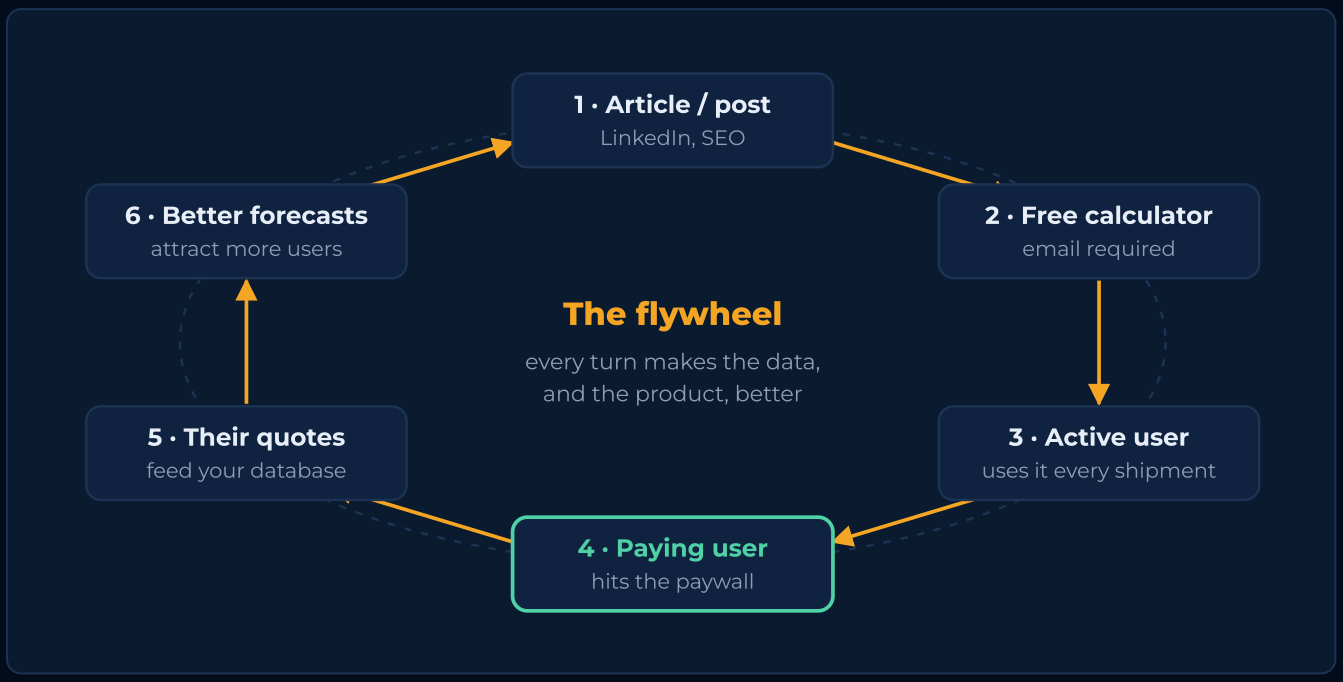
1. **New prices come in**
   You enter SCFI and WCI in the admin panel, or a script fetches them.
2. **Real quotes are added**
   Your own, and the ones users submitted this week.
3. **Algeria premium updated**
   Per port: Algiers, Oran, Djen Djen...
4. **Forecasts recalculated**
   12 weeks ahead, for every route.
5. **Alerts sent**
   Users following a route that moved more than X% get an email.

WHAT LIVES IN THE DATABASE

TABLE	WHAT'S INSIDE	FILLED BY
Rates	Freight price per route, per week	Friday robot + you
Forecasts	12-week prediction for each route	Friday robot
Duty bands	Duty and VAT % per product type	You (admin panel)
Risk rules	"If this + that, then warn" list	You (admin panel)
Users	Email, plan, routes they follow	Sign-up
Calculations	Every calculation, anonymised	The brain
Quotes	Real forwarder quotes, by port and week	Users + you

4 The money machine

How a stranger becomes a paying user, and why each user makes the product better for the next one.



FREE (AFTER EMAIL)

- Engine A · landed cost
- Engine C · quote check
- Engine D · risk flags

Enough value to use before every shipment. This is what brings people back.

PAID (MONTHLY)

- Engine B · forecast, Book now / Wait
- Price alerts on their routes
- Saved shipments and history
- PDF report for their boss
- Team accounts (later)

The 6 numbers to track every week

1	2	3	4	5	6
Visitors	**Signups**	**Active**	**Paying**	**Churn**	**MRR**
people who land on the site	gave their email	did at least one calculation	subscribed	% who cancel each month	monthly recurring revenue

If a number drops, the problem is in the step just before it. Few signups means a weak landing page. Few paying users means the paid features aren't valuable enough yet.

5 Build it in 3 stages

Don't build everything at once. Each stage must prove something before you pay, in time or money, for the next one.

STAGE 0 · START HERE

Lead magnet

about 1 week

GOAL	Collect emails and real quotes. Prove people use it.
TOOLS	Webflow page + calculator embedded + Beehiiv email form + Google Sheet as the database
NOT YET	No login, no payment, no server. Formulas are visible, which is fine for a free tool.
COST	Almost zero: tools you already have
DONE WHEN	**50 people used it and 20 real quotes collected**

STAGE 1

Real SaaS

3–6 weeks

GOAL	First paying users.
DATABASE + LOGIN	Supabase (free tier)
APP SCREEN	Web app built with Claude Code, hosted on Vercel (free tier), at app.yourdomain.com
WEEKLY ROBOT	Python script on GitHub Actions (free scheduled runs). The forecast model lives here.
PAYMENT	See the box below. Research this first.
DONE WHEN	**10 paying users**

STAGE 2

Scale

after 50+ paying users

ADDS	Team accounts, route alerts, Level 2–3 forecast, quote benchmark per port, forwarder scorecards
DONE WHEN	**Revenue covers the time you spend on it**

PAYMENT: THE BLOCK TO SOLVE FIRST

Stripe doesn't accept Algerian businesses. Options to check:

- **Algerian clients:** an Algerian payment gateway like Chargily Pay (Edahabia and CIB cards), or bank transfer with an invoice for B2B.
- **International clients:** a reseller service such as Paddle or Lemon Squeezy. Check whether they accept Algerian sellers before planning around them.

This decides how you get paid, so it comes before any Stage 1 code.

6 Who does what

You need to understand the blocks, not write the code.

YOU · THE MOAT

- The data: rates, quotes, duty bands
- The risk rules from your field experience
- Selling: LinkedIn, articles, DMs
- Deciding what's free and what's paid
- Talking to users every week

CLAUDE · THE BUILD

- Writing all the code
- Database design
- The forecast model and backtest
- Hosting and the weekly robot
- Fixing and improving as you grow

YOUR NEXT 7 DAYS

- [ ] **Write the rate table for 6 Algerian ports** from quotes you already see at work.
- [ ] **Write 20 risk rules** "If Incoterm X + payment Y, then warn Z." This is your expertise in writing.
- [ ] **Start the Friday tracking sheet** SCFI, WCI, your Algiers quotes, GRIs announced.
- [ ] **Decide the free/paid line** Start with the split on page 4.
- [ ] **Check payment options** What Chargily (or another gateway) needs to open an account.
- [ ] **Claude builds Stage 0** the calculator live on your site, with the email gate.

Words you'll hear

Frontend	What the user sees and clicks (the app screen).
Backend / API	The server side that does the work and answers the screen (the brain).
Database	Where all the information is stored in tables.
Cron job	A task that runs automatically on a schedule (the Friday robot).
Paywall	The point where free features stop and the user must subscribe.
Free tier	The free plan of a tool like Supabase or Vercel, enough to start.
MRR / Churn	Monthly recurring revenue / % of subscribers who cancel each month.

Tool names and free tiers are current suggestions. Check pricing and country availability before committing.